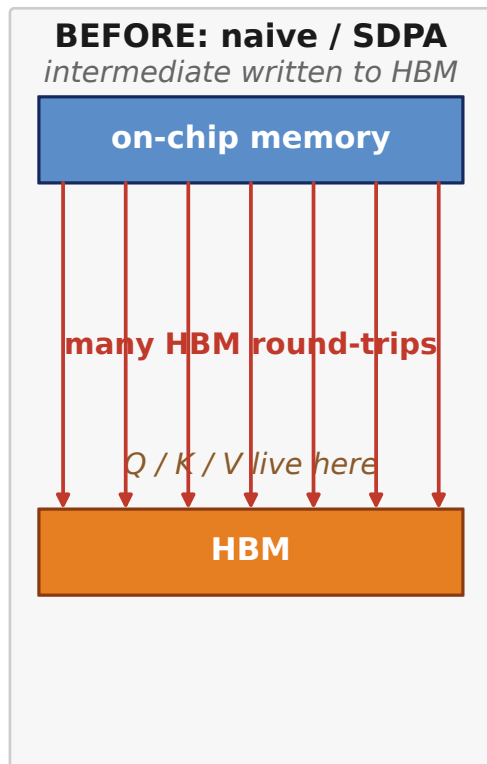
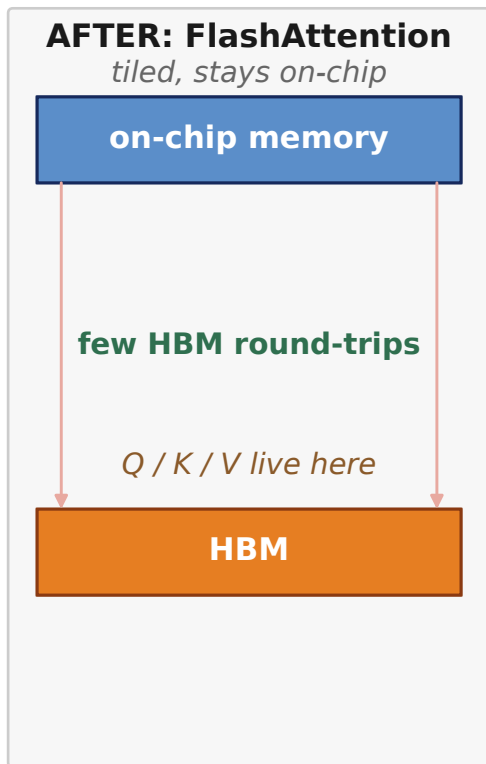


FlashAttention: the same math, far less HBM traffic



$O(L^2)$ HBM traffic:
every query re-reads the row-block



$O(L)$ HBM traffic:
each tile read once, softmax online

**same $Q \cdot K^T \cdot V$ math and same result —
the gain is fewer HBM round-trips, not cheaper attention**